

ELEY: A Hardware-Readiness Perspective on Industrial Robot Learning

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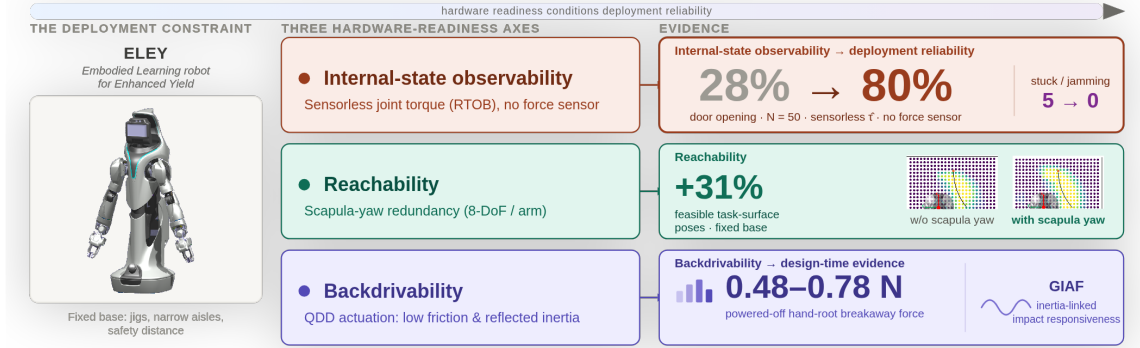


Fig. 1: Overview of ELEY and the hardware-readiness framing. Fixed-base constraints motivate three body properties: task-relevant reachability, backdrivability, and internal-state observability. Scapula yaw increases feasible task-surface poses by 31%; powered-off hand-root breakaway forces range from 0.48 N, with GIAF on a same-QDD testbed providing complementary inertia-linked evidence; sensorless torque observation raises door-opening success from 28% to 80%.

Abstract—Robot learning is approaching industrial deployment, yet factory-floor reliability is usually framed as a question of policies and data. We ask a complementary question: whether the robot embodiment is ready for learning-based deployment. Using ELEY, a bimanual mobile manipulator, we organize hardware readiness along three axes: task-relevant reachability, backdrivability, and internal-state observability. Adding one scapula-yaw joint to each arm increases feasible poses on a fixed-base door-opening task surface by 31%. On the full ELEY hardware, powered-off hand-root breakaway forces range from 0.48 to 0.78 N across the tested arms and directions; complementary GIAF analysis on a same-QDD-configuration testbed characterizes inertia-linked impact responsiveness. Finally, adding sensorless reaction-torque estimates to the proprioceptive observation of a fine-tuned vision-language-action (VLA) policy raises door-opening success from 28% to 80% over 50 trials per condition. These results suggest that robot embodiment and internal-state observation should be treated as primary design variables for reliable industrial robot learning.

I. INTRODUCTION

Robot learning is approaching industrial deployment, and whether it meets the precision, reliability, and safety standards of the factory floor is usually framed as a question about policies and data. We ask a complementary question: whether the robot embodiment is ready for learning-based deployment. We argue that *hardware readiness* is a primary design factor that conditions deployment reliability, alongside the model and the data. Here, deployment reliability is distinct from system-level operational safety: emergency

stops and power cut-off protect the platform, whereas hardware readiness concerns whether the body and observations allow a learned policy to execute contact-rich tasks robustly under constraints.

Industrial contact tasks rarely occur in open space. The mobile base is routinely constrained by jigs, fixtures, narrow aisles, and safety distances, and in bimanual operation the feasible set of end-effector poses shrinks further under arm interference and coordination constraints. Under such constraints, three properties of the robot body—often treated as fixed—directly shape whether a learned policy can be demonstrated, deployed, and trusted: (i) *reachability*, the availability of task-required end-effector poses when the base cannot move; (ii) *backdrivability*, the passive mechanical responsiveness that governs how safely and promptly the end effector reacts to contact; and (iii) *internal-state observability*, the access to contact-relevant signals such as joint torque that vision and joint angles alone cannot disambiguate. We refer to these three properties collectively as hardware readiness.

We study hardware readiness on ELEY (Embodied Learning robot for Enhanced Yield), a bimanual mobile manipulator developed as part of a broader effort toward learning-enabled industrial manipulation that spans low-inertia arm design [1] and large-scale data collection. Each arm adds a scapula-yaw degree of freedom at its base, forming an 8-DoF redundant configuration that expands task-relevant reachability under a fixed base. All joints use quasi-direct-drive (QDD) actuation [2] to reduce friction and reflected inertia, improving backdrivability. Joint reaction torque arising from environmental interaction is estimated without torque sensors

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via a reaction torque observer (RTOB) [3] and exposed as a contact-state cue. A vision-language-action model, π_0 [4], is fine-tuned on teleoperated demonstrations and deployed on the real robot.

We report evidence for the value of hardware readiness on two full-sequence, contact-rich tasks executed on real hardware: door opening, chosen to stress geometric constraint mismatch, and peg hammering, chosen to stress impulsive loading. The former’s dominant failure mode—jamming under a hinge constraint—is shared with industrial insertion and machine-tending tasks such as operating machine doors, and we use it to isolate the effect of internal-state observability under an otherwise identical policy.

Contributions.

- 1) We frame task-relevant reachability, backdrivability, and internal-state observability as hardware-readiness requirements for fixed-base, contact-rich industrial robot learning.
- 2) We provide design-time evidence through task-surface reachability and full-platform breakaway-force measurements on ELEY, complemented by inertia-linked GIAF analysis on a same-QDD-configuration testbed.
- 3) We perform a controlled real-robot ablation showing that adding sensorless torque observation raises fixed-base door-opening success from 28% to 80%.

II. RELATED WORK

Internal-state observation for contact-rich learning. Contact-rich manipulation has long benefited from force and torque feedback, from impedance and operational-space control [5], [6] to recent force-aware policy models [7]; surveys of peg-in-hole assembly [8] and of imitation learning for contact-rich tasks [9] document this consistently. Sensorless joint-torque estimation is likewise well established, from the reaction torque observer [3] to backdrivable QDD arms that expose current-based torque estimates for learning and teleoperation [10]. Our aim is therefore not a new sensing mechanism but a deployment-level question: we treat the *observation space* as a hardware-readiness choice and quantify, under an otherwise identical learned policy, how much a sensorless estimate—in contrast to the dedicated force/torque or tactile sensing assumed by force-aware policies [7]—already improves reliability.

Embodiment interfaces for data and deployment. Recent robot-learning systems increasingly treat embodiment and data-collection interfaces as part of the policy-training stack, from low-inertia backdrivable arms [1] to YUBI-style large-scale bimanual demonstration interfaces [11]. While such work primarily improves data acquisition and operator throughput, we ask the complementary deployment question: which body properties make human-collected end-effector trajectories executable and reliable under fixed-base industrial constraints?

Task-relevant reach and safety context. Manipulability [12], [13] and capability/reachability maps [14], [15], [16] characterize where a manipulator can act, often to plan mobile-base placement at run time. Under fixed-base

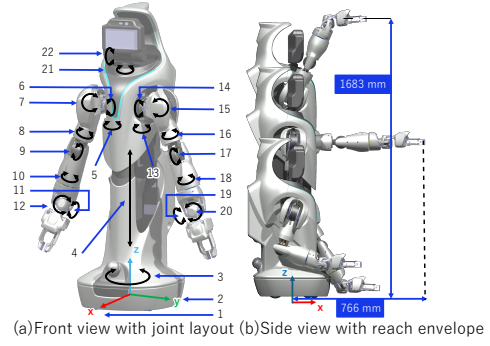


Fig. 2: ELEY has two 8-DoF arms (each adding a scapula-yaw joint, highlighted), a 3-DoF omnidirectional base, a 1-DoF lifting torso, and a 2-DoF head. Horizontal reach extends to 766 mm, with the end effector kept horizontal at heights from 100 to 1680 mm.

TABLE I: Key specifications of ELEY.

Item	Specification
Mobile base	Omnidirectional (offset differential drive, two passive casters)
Torso lift	Stroke 680 mm (shoulder height 500–1180 mm)
Single arm	8-DoF (scapula + 7-DoF); reach 700 mm; all QDD joints
Dimension	Footprint 430 mm; height (min/max) 940/1620 mm
Gripper	Swappable; ISO 9409-1 compliant
Sensing	Head/hand RGB-D, rear RGB, front/rear 3D LiDAR; proprioception $q, \dot{q}$ (logged); sensorless torque estimate $\hat{\tau}$ (RTOB)
Control	Inner current/RTOB update 62.5 kHz; joint servo 500 Hz (ROS 2 + EtherCAT); policy command 10 Hz
Safety	Emergency stop; ECU monitoring; power cut-off

bimanual conditions, however, interference and coordination constraints can sharply restrict feasible poses; we therefore use feasible-pose count over an environment-fixed task surface as a design-time readiness metric. Prior work on intrinsically safe platform design [17] and collision detection and reaction [18], [19] addresses complementary system-level safety mechanisms; here we focus on whether reachability, passive contact response, and internal-state observation make learned contact-rich deployment reliable on a single platform.

III. DESIGN-TIME HARDWARE READINESS OF ELEY

ELEY has an omnidirectional base, a lifting torso, a 2-DoF head, and two 8-DoF arms (22 DoF excluding hands; Fig. 2, Table I). This section presents design-time evidence for reachability and backdrivability; deployment-level evidence for internal-state observability is presented in Sec. IV. We call the additional yaw DoF at each arm base *scapula yaw*.

A. Reachability: task-relevant poses under a fixed base

Requirement. When the base cannot move, the arms must supply task-required poses while avoiding joint limits, self-collision, and bimanual interference. Additional task-relevant reachability provides post-grasp posture slack after contact engagement.

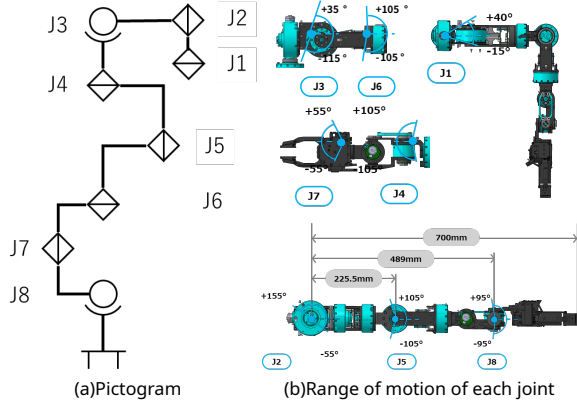


Fig. 3: Arm kinematics and joint ranges. The highlighted scapula-yaw joint is added proximally to a conventional 7-DoF arm.

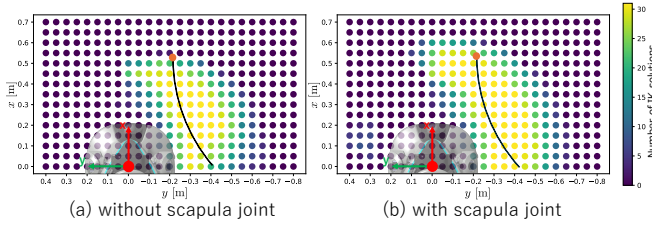


Fig. 4: Task-relevant reachability on the door-opening surface with base and torso fixed. Heatmap: feasible orientations per grid point, without (left) vs. with (right) scapula yaw. Red: robot center; orange: knob; curve: knob trajectory. Feasible poses: 2365 to 3098 (+31%).

Design. Each arm adds a proximal scapula-yaw joint to a conventional 7-DoF structure, forming an 8-DoF redundant arm (Fig. 3). This expands posture options without adding mass to distal links. Scapula-like redundant arms have been explored before [20]; here proximal redundancy is treated as a design-time readiness variable rather than a new mechanism.

Evidence. We evaluate the plane swept by the grasped end effector during door opening, with base and torso fixed. We sweep $x \in [0.0, 0.7]$ m and $y \in [-0.8, 0.4]$ m at 0.05 m resolution, $z = -0.25$ m from the shoulder, and roll/pitch/yaw in $[-50^\circ, 50^\circ]$ at 10° resolution. Feasible IK solutions satisfy joint limits and self-collision constraints. Compared with scapula yaw fixed at 0° , the movable case increases feasible poses from 2365 to 3098 (+31%; Fig. 4). This is a readiness margin rather than a task-performance claim. In a jointly constrained bimanual case, feasible poses increase from 220 to 324 (+45%; Fig. 5).

B. Backdrivability: passive contact responsiveness

Requirement. Passive end-effector response depends on drivetrain friction and reflected inertia. High friction increases the force required to initiate motion, whereas high reflected inertia limits impact accommodation. We therefore assess these two components separately as design-time indicators of backdrivability.

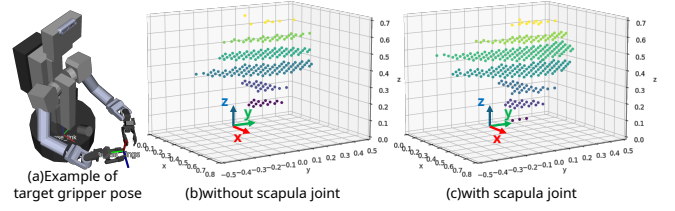
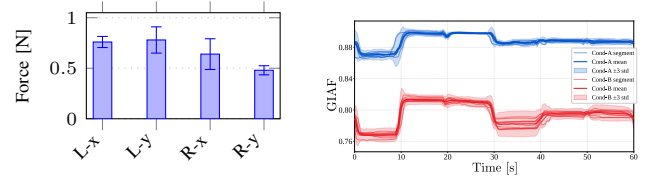


Fig. 5: Jointly constrained bimanual case without (left) and with (right) scapula yaw. Feasible poses increase from 220 to 324 (+45%).



(a) Breakaway force on ELEY.

(b) Inertia-linked GIAF.

Fig. 6: Design-time evidence for backdrivability. (a) Mean $\pm$ sample SD powered-off breakaway force at the ELEY hand root with short-circuit braking disabled ($N = 5$ per condition). (b) GIAF on a fixed-base, same-QDD-configuration testbed: baseline QDD vs. increased apparent inertia.

Design. All ELEY joints use QDD actuation [2], [1], with gear ratios kept as low as the required torque permits to reduce drivetrain friction and reflected inertia.

Evidence. We first evaluate friction-related backdrive resistance directly on ELEY. With the robot powered off and short-circuit braking disabled, a force gauge was applied quasi-statically to the hand-root bracket in a fixed reference posture until motion onset. Across five trials per condition, mean breakaway forces were 0.76 and 0.78 N for the left arm and 0.64 and 0.48 N for the right arm in the tested fore-aft and lateral directions, respectively (Fig. 6a). For the inertia-related component, we use the generalized impact-absorption factor (GIAF) [21], [22] as a design-time proxy; larger values indicate a smaller rotor-inertia penalty on task-space impact response. On a fixed-base testbed sharing ELEY’s all-QDD joint configuration, the baseline QDD condition exhibits higher GIAF than a condition with increased apparent inertia (Fig. 6b), isolating the arm-side inertia effects considered in this analysis. These results support low backdrive resistance and low-inertia contact response, but are not a full-platform dynamic guarantee or a causal explanation of door-opening success.

IV. INTERNAL-STATE OBSERVABILITY IN LEARNED DEPLOYMENT

Observation design. In contact-rich tasks, constraint engagement, sticking, and insufficient pressing may be difficult to infer from RGB images and joint angles alone. ELEY exposes sensorless RTOB estimates as an additional proprioceptive input. The estimates are synchronized to the 10 Hz policy input and normalized using training-set statistics. We

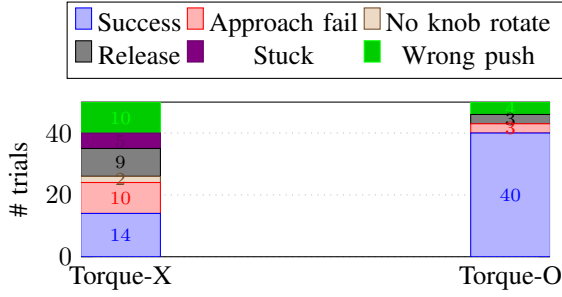


Fig. 7: Door-opening outcomes by dominant failure mode ($N = 50$ per condition). Stuck/jamming drops from 5 (Torque-X) to 0 (Torque-O).

use $\hat{\tau}$ as a relative cue for contact transitions; quantitative accuracy, observer bandwidth, and posture dependence remain to be characterized.

Policy and controlled ablation. We fine-tune π_0 [4] on teleoperated door-opening demonstrations using RGB streams from the head and hand cameras and a fixed language prompt; depth is not provided to the policy. The policy action is a vector of joint-position targets at 10 Hz, not Cartesian, force, or torque commands. The full observation is

$$o_t = (I_t^{\text{head}}, I_t^{\text{hand}}, l, p_t),$$

where l is the fixed language prompt and p_t is the proprioceptive input. We denote the torque-observed and torque-excluded policies as Torque-O and Torque-X, respectively, and compare

$$p_t = \begin{cases} [q_t^\top, \hat{\tau}_t^\top]^\top & \text{Torque-O,} \\ [q_t^\top]^\top & \text{Torque-X.} \end{cases} \quad (1)$$

Both policies use the same 110 demonstrations ($\sim 40k$ steps). Torque-X removes only $\hat{\tau}$ during training and inference; visual inputs, language prompt, backbone, action space, normalization, training schedule, and safety mechanisms are unchanged.

Real-robot evidence. Door opening is constraint-dominant: after grasping the knob, motion is constrained by the hinge, and small pose or friction errors can cause jamming or grasp breakdown. Before each trial, small perturbations were manually applied to the initial arm posture. In addition, the mobile base was not reset between trials, resulting in small natural variations in the initial base pose, while the door geometry remained unchanged. Both policies were evaluated using the same protocol ($N = 50$ per condition, 60 s timeout); success required fully opening the door. Torque observation raises success from 28% (14/50) to 80% (40/50), with Fisher’s exact test $p = 2.9 \times 10^{-7}$. Fig. 7 shows reductions across multiple failure modes; notably, stuck/jamming drops from 5 Torque-X trials to 0 Torque-O trials. This pattern is consistent with $\hat{\tau}$ providing an additional cue during pulling and pushing, where incipient constraint mismatch is difficult to disambiguate visually. This complements force-aware policies relying on dedicated

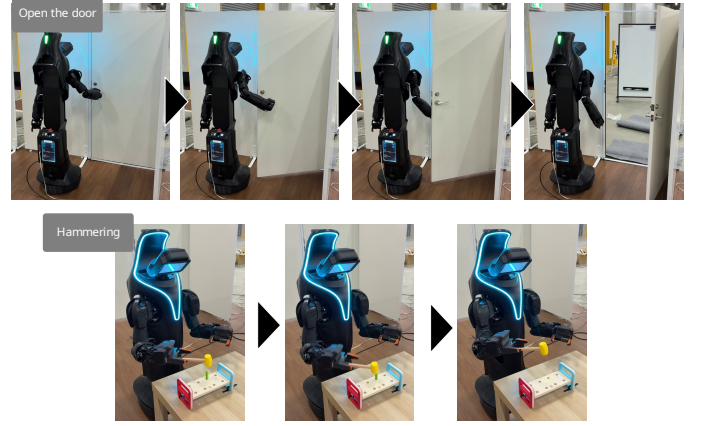


Fig. 8: Task execution snapshots. Top (left-to-right): door opening (approach, pull, reach-around, push to fully open). Bottom (left-to-right): hammering (backswing, impact, completion).

sensing [7]; phase- and joint-level attribution remains future work.

As an additional impulse-dominant feasibility demonstration, ELEY completed peg hammering in 20/20 Torque-X trials; this is not an observability ablation. Representative execution sequences for door opening and hammering are shown in Fig. 8. Door opening includes approach, pulling, reach-around, and pushing phases, while hammering shows backswing, impact, and completion.

V. DISCUSSION AND CONCLUSION

The evidence levels in this paper are intentionally different. Reachability and backdrivability are design-time readiness measurements: scapula yaw provides geometric slack, while full-platform breakaway-force measurements and inertia-linked GIAF analysis characterize low-resistance passive response. Internal-state observability provides the strongest deployment-level evidence through a controlled real-robot ablation. The backdrivability results should not be interpreted as a causal explanation of the policy improvement; rather, low-friction, low-inertia actuation supports passive accommodation during contact, while $\hat{\tau}$ gives the policy an additional cue for contact transitions. Likewise, the 28% to 80% result does not imply that sensorless torque universally replaces dedicated force/torque sensing, but shows that an internal hardware signal can substantially improve reliability in this constraint-dominant regime.

Limitations. The study covers one environment and does not evaluate continuous force-regulation tasks. The breakaway-force result is posture- and direction-dependent, GIAF is evaluated on a fixed-base, same-QDD-configuration testbed, and full-platform apparent-mass identification remains future work. Quantitative RTOB accuracy, bandwidth, and phase/joint attribution also remain open.

Overall, ELEY supports a hardware-readiness view of industrial robot learning: proximal redundancy expands design-time feasible poses, QDD actuation provides low backdrive

resistance and inertia-linked responsiveness, and sensorless internal torque observation raises fixed-base door-opening success from 28% to 80%.

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